

# Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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**Abstract**—Artificial-intelligence (AI) data centers can provide demand response (DR), but a meter-only baseline cannot distinguish a realizable workload shift from strategic deferral, rebound, or a transmission-feasible response. We present a security-aware workload-state verifier and nodal settlement that link a committed job ledger, feasible service trajectories, and contingency-constrained network value. A sparse multi-period linear program enforces release, conservation, deadlines, capacity, migration, and cross-midnight completion. A validation-only convex quadratic program with a linear false-credit/CVaR epigraph and a two-sided event band gives an auditable risk contract; an event-gate mode never credits unsubmitted future work. Signed dispatch-value differences under finite non-landing single-contingency (N-1) constraints determine payments, with exact coalition accounting for allocation. On 5,188,507 BurstGPT requests and 71,128 positive-energy MIT Supercloud jobs, the risk-constrained verifier reduces unsupported credit from 1.849211 to 1.826322 MWh/day (1.24%) relative to the selected single feasible cap, while its nRMSE changes from 0.345209 to 0.345019. Relative to the feasible-quantile comparator, unsupported credit falls by 37.40% (2.917471 to 1.826322 MWh/day); exact three-day block tests are reported without claiming uniform accuracy improvement. Exact N-1 settlement reduces trace-anchored error from \$165.288 to \$1.018/day across 37 RTS-24 outages, while end-to-end errors are \$192.978, \$159.514, and \$159.483/day for base-case exact, N-1 linear, and N-1 exact settlement. Independent replay covers every locked 15-minute slot and job-level residuals are below  $10^{-17}$  MWh. Held-out conversion endpoints, four public AC networks, spatial permutations, and a workload-hull payment interval test the declared uncertainty boundary. The non-co-located public traces define a reproducible benchmark, not a utility field event.

**Index Terms**—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

## I. INTRODUCTION

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch GPU work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value.

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice, despite known bias and manipulation incentives [1], [2]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [1]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine

whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide the missing physical state. Temporal deferral, dynamic capacity, and geographical load balancing can reduce energy cost and peak demand [3]. Geographical assignment provides the complementary spatial control mechanism [4]. Recent work assesses data-driven flexibility and aggregator coordination [5], [6], while multi-level dispatch refines delay-sensitive electrical response. Hierarchical optimal-consumption models further connect facility efficiency with response quality [7]. Virtual-link market clearing and space-time remuneration establish how geographical and temporal transfers can enter electricity markets [8], [9]. Receding-horizon operation and uncertainty-aware frequency-regulation coordination further establish the need to retain future arrivals and multi-site state [10]. However, these methods optimize an operator-controlled schedule rather than verify an independently submitted DR counterfactual against an auditable workload ledger. This separation between computing feasibility and market verification motivates the auditable ledger used here.

The second gap is electrical. Geographic workload control is commonly valued with energy prices or gross megawatts. Such accounting ignores the fact that nodal marginal value changes when congestion or a contingency constraint binds. Direct-current (DC) power flow and linear distribution factors provide transparent network sensitivities [11], [12]; N-1 dispatch can impose every credible post-contingency line limit through line-outage distribution factors (LODFs) [13]. No existing data-center DR verification framework jointly certifies workload feasibility, preserves signed spatial response, and settles the resulting security-aware network value. Public workload and facility telemetry can support an observational replay of that ledger, but it does not constitute a causal utility event; a rigorous study must therefore separate measured-execution replay, counterfactual construction, and network-value verification rather than present them as one field intervention.

This paper makes three power-system contributions. 1) We formulate a provenance-bound workload-state demand model in which release, deadline, conservation, cross-midnight completion, capacity, and migration constraints are enforced simultaneously. The resulting job-level witness links each joined job to its immutable scheduler/DCGM record and preserves measured execution energy. 2) We construct a validation-only, continuous-arrival counterfactual verifier with a pointwise credit envelope. The output is an attainable spatial-temporal load trajectory rather than an unconstrained meter-only prediction, and its statistical performance is evaluated under an explicit false-credit/under-credit tradeoff. 3) We develop signed space-time nodal remuneration and a bilateral net-value contract under preventive N-1 dispatch, and map the feasible workload set into a payment uncertainty interval over

held-out workload-to-power conversion. The novelty is the coupling invariant that carries one committed workload state into physical verification, network valuation, and a set-valued settlement rule; the individual LP, dispatch, and allocation primitives are standard and are not claimed as new in isolation. The certificate is conditional on trusted scheduler and telemetry provenance: a cryptographic digest verifies record integrity after commitment but does not authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

## II. SYSTEM MODEL AND PROBLEM FORMULATION

### A. Spatial workload state

Let  $\mathcal{S}$ ,  $\mathcal{K}$ ,  $\mathcal{D}$ , and  $\mathcal{T}$  denote source regions, workload classes, destination data centers, and 15-minute intervals. The measured ledger supplies arrival energy  $a_{skt}$  for source  $s$ , class  $k$ , and release interval  $t$ . The decision  $x_{skd\tau} \geq 0$  is the energy served at destination  $d$  in interval  $\tau$ . Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^t \sum_{d \in \mathcal{D}} x_{skd\tau}. \quad (1)$$

With cumulative arrivals  $A_{skt} = \sum_{\tau=0}^t a_{sk\tau}$  and deadline  $D_k$ , release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \quad (2)$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \quad (3)$$

The state recursion equals service, terminal service equals total arrivals, and  $\sum_{s,k} x_{skd\tau} \leq \bar{P}_d \Delta t$ . Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skd\tau}. \quad (4)$$

For rolling window end  $T^+$  and final event-day release  $T^0$ , the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk, \max\{T^0, T^+ - D_k\}}, \quad (5)$$

which completes every job due inside the window and every event-day arrival, while retaining later real arrivals in the release constraints. The window starts  $D_{\max} + 96$  intervals before the event day and ends  $D_{\max}$  intervals after it; omitted earlier work must therefore expire before the evaluated day. Equations (2) and (3) prevent early execution and late completion, while (4) maps the feasible service state to the facility meter. This model follows workload deferral and geographical assignment foundations [3], [4]; its rolling implementation follows the state-retention principle in [10]. Waiting and migration costs are  $w_k(\tau - t)x_{skd\tau}$  and  $m_{sd}x_{skd\tau}$ , respectively. Because arrivals are fixed, replacing waiting duration by service time adds only a constant.

For provenance, each retained job is represented by the canonical tuple and its digest in

$$\begin{aligned} \ell_j &= (\text{id}_j, t_j^{\text{submit}}, t_j^{\text{start}}, t_j^{\text{end}}, E_j, n_j^{\text{tel}}, q_j, \kappa_j, \sigma_j), \\ \mathcal{L} &= \text{sort}\{\ell_j : j \in \mathcal{J}\}, \\ h_{\mathcal{L}} &= \text{SHA256}(\text{serialize}(\mathcal{L})). \end{aligned} \quad (6)$$

where  $E_j$  is the positive aggregated Data Center GPU Manager (DCGM) energy and  $n_j^{\text{tel}}$  is its raw telemetry-record count. The digest uses the Secure Hash Standard [14]; the canonical row

order, one-to-one join, release–execution order, and raw-to-join energy equality are checked independently of the scheduling linear program (LP). Consequently, the flow witness certifies the exact committed ledger and measured profile, while the digest does not make an unverified claim about jobs that might have been submitted before the commitment. Experiment 16 reports this provenance certificate and reconciles the measured regional capacity envelope with the 118-MW nameplate that was committed before the validation/test split; the envelope is not used to select that nameplate.

The trust boundary is explicit. The verifier assumes that the scheduler and DCGM services are the authoritative sources for the fields used in  $\ell_j$ , and that the digest is committed before the event decision. Under this assumption, a changed record is detectable and a joined record cannot be silently replaced after commitment. The digest does not, however, establish that a semantically false deadline, region label, service class, or energy value was truthful when it was signed. Authenticity of those fields requires an upstream scheduler attestation, meter signature, or trusted execution interface; it is outside the optimization certificate proved here.

### B. Counterfactual, response, and false credit

For event set  $\mathcal{E} \subset \mathcal{T}$ ,  $p_{dt}^0$  denotes the no-event counterfactual and  $p_{dt}^1$  denotes the event trajectory generated by the declared workload optimization. The mechanism-isolation experiments use  $p_{dt}^{1, \text{sim}}$  for this optimized response, whereas Experiment 13 uses  $p_{dt}^{1, \text{meas}}$  for the independently measured MIT/DCGM execution trace. We reserve  $p_{dt}^{\text{obs}}$  for measured execution used only as scoring truth; it is never supplied to an estimator or used to claim a causal utility event. For either response trace,  $r_{dt} = p_{dt}^0 - p_{dt}^1$  is signed: gross local response is  $[r_{dt}]_+$ , while net response retains negative rebound. For an estimate  $\hat{p}^0$ , unsupported credited energy is

$$F = \Delta t \sum_{d,t \in \mathcal{E}} [\hat{p}_{dt}^0 - \max\{p_{dt}^{\text{obs}}, p_{dt}^1\}]_+. \quad (7)$$

The audit computes (7) separately for every locked day so that a favorable aggregate error cannot hide unsupported positive credit. We also report normalized root-mean-square error (nRMSE), bias, precision, recall,  $F_1$ , under-credit, payment error, and overpayment.

### C. Strategic reference scheduling

Consider an arithmetic mean of  $H = 10$  reference days, event probability  $q$ , and DR price  $\pi^{\text{DR}}$ . If reference day  $j$  adds event-window service  $\delta R_j$ , the expected payment change is

$$\Delta \Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^H \delta R_j - \sum_{j=1}^H \Delta C_j. \quad (8)$$

Let  $c'_j$  be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day  $j$ . Linear-program sensitivity [15] gives the following exact local condition:

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \quad (9)$$

The historical-baseline premise is documented in [1]; (8)–(9) are this paper’s workload-specific specialization.

### D. Network value

Let  $M$  be the data-center-to-bus incidence matrix and  $p_t = p_t^{\text{native}} + M p_t^{\text{dc}}$  the nodal demand vector. For generator-to-bus

map  $C_g$ , generator bounds, power transfer distribution factor (PTDF)  $H$ , and line ratings  $\bar{f}$ , the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \sum_i C_i(g_i) \\ \text{s.t. } \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \leq g \leq \bar{g}, \\ -\bar{f} \leq H(C_g g - p) \leq \bar{f}. \end{aligned} \quad (10)$$

The incidence mapping aggregates device demand at its connection bus, and (10) is the standard lossless DC representation [11], [12]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \quad (11)$$

Positive  $\Delta V$  is a credit and negative  $\Delta V$  a debit; no positive-only truncation is applied.

### III. PROPOSED METHODOLOGY

#### A. Framework and matched-information pre-estimation

Fig. 1 shows the complete chain. Meter history and the submitted-job ledger enter a statistical pre-estimator and the workload feasible set in parallel. The pre-estimator never receives execution energy used as scoring truth. For each event day, it uses only the information available under one of two declared protocols: lagged meter/calendar features for online baselines, or the complete submitted ledger for equally informed post-event comparisons. Ridge regression and gradient boosting follow [16], [17]; extremely randomized trees follow [18]. Two stronger comparators are also solved: median-loss gradient boosting with the complete submitted ledger and a nonnegative synthetic control whose weights globally minimize pre-event absolute error over 28 donor days [19]. To separate the effect of projection from that of pre-estimation, the complete-ledger quantile output is also passed through the same exact workload-feasible projection, with its penalty chosen by contiguous validation folds. The same complete ledger is supplied to the post-event learner, the selected single projection, and the proposed convex verifier.

#### B. Sparse cumulative workload-state verifier

For statistical estimate  $\tilde{p}$  and each predeclared penalty  $\rho_\ell$ , the first-stage program solves

$$\min_{x,y,z} C_w(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \geq \pm(p_{dt}(x) - \tilde{p}_{dt}), \quad (12)$$

subject to (1)–(4). Absolute-value epigraphs are standard convex constructions [15]; using them to project a meter baseline onto an auditable workload ledger is specific to this work. Here  $C_w$  denotes workload operating cost;  $C_i$  denotes generator-segment cost in (10), and  $V(p)$  denotes the network dispatch value. Thus (12) changes the objective, not the physical feasible set. Six globally solved schedules  $p^{(\ell)}$  share identical arrivals and constraints. Their validation-only convex combination  $\bar{p} = \sum_\ell \alpha_\ell p^{(\ell)}$ ,  $\alpha \geq 0$ ,  $\mathbf{1}^\top \alpha = 1$ , remains feasible because the common scheduling set is convex.

The coefficients minimize validation squared error subject to both total and daily-tail false-credit budgets, where conditional value at risk (CVaR) measures the latter [20]. Let  $\ell^*$  be the single candidate with minimum worst contiguous-fold nRMSE and  $B_j^{\text{cap}}$  its validation false exposure on day  $j$ . For reserve  $\beta \in (0, 1]$ ,

$$\sum_j F_j(\alpha) \leq \beta \sum_j B_j^{\text{cap}}, \quad (13)$$

$$\text{CVaR}_{0.75}(F_j(\alpha)) \leq \beta \text{CVaR}_{0.75}(B_j^{\text{cap}}). \quad (14)$$

For implementation, the positive part is not differentiated through a sort. For each event sample  $n$  and validation day  $j$ , the program introduces  $f_n \geq 0$ , a CVaR threshold  $\nu \geq 0$ , and tail slacks  $u_j \geq 0$ :

$$f_n \geq p_n^\alpha - \max\{p_n^0, p_n^1\}, \quad F_j = \sum_{n \in j} f_n, \quad (15)$$

$$F_j - \nu - u_j \leq 0, \quad (16)$$

$$\nu + \frac{1}{0.25J} \sum_j u_j \leq \beta \text{CVaR}_{0.75}(B_j^{\text{cap}}). \quad (17)$$

Together with the simplex equality, box bounds on  $\alpha$ , and the total budget in (13), these are linear constraints and the squared error objective is a convex quadratic program. A HiGHS feasibility solve is followed by a sparse trust-region solve; the reported KKT and primal residuals are below  $10^{-8}$  in every fitted fold. Four contiguous folds select  $\beta$  using validation nRMSE, then risk ratio; held-out noninferiority is reported as an outcome and is not used as a hidden constraint. A second exact projection targets  $\bar{p}$  while imposing, for every event sample,

$$p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}. \quad (18)$$

a) *Two-sided credit band.*: The upper envelope alone would admit an arbitrarily conservative zero-credit profile. We therefore impose the matching lower band on the event slots,

$$p_{dt}^{(\ell^*)} - \varepsilon \leq p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}, \quad (d, t) \in \mathcal{D} \times \mathcal{E}, \quad (19)$$

where  $\varepsilon = 1$  MW is fixed before the locked test split. The lower bound is implemented together with the workload constraints, not as a post-processing clip. Thus the output remains an attainable schedule while its credited event trajectory is within a declared band of the selected single projection. The total constraint (13) controls aggregate exposure and (14) controls the worst daily tail; the pair (18)–(19) supplies the samplewise credit certificate. Since  $[z - c]_+$  is monotone, the upper inequality bounds false credit for any unseen actual trajectory, while the lower inequality bounds the additional under-credit by at most  $\varepsilon|\mathcal{D}||\mathcal{E}|\Delta t$  relative to the selected cap. The selected single feasible projection proves nonemptiness, and the locked results report both margins explicitly.

b) *Decision-time information boundary.*: The ex-post selector can use the complete submitted ledger because it is an offline settlement protocol. For an event-gate deployment, all arrivals after the gate are removed before both the causal target construction and optimization, and terminal completion is required only for work due by the gate. The target is refit with the same metadata gradient-boosting specification used by the matched-information baseline, using historical days unchanged and the current day's masked ledger for every feature row. We additionally solve a committed-ledger safe mode with  $p_{dt} \leq P_d^{\text{fix}}$  on the event slots. This mode is an exact workload LP, uses no future arrivals, and credits only service backed by the committed ledger; later arrivals become eligible only after commitment. It is reported separately from the information-boundary diagnostic and the ex-post benchmark.

The cumulative state reduces the constraint-matrix nonzeros from  $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|^2)$  for explicit prefixes to  $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$ . A 512-slot deadline is handled by (5) with every observed future arrival retained; no zero-arrival completion buffer is used in the final continuous-horizon certificate. The day trajectory is embedded by a lexicographic pair of linear programs: minimum L1 distance first, followed by minimum operating cost on that optimal-distance face. Hence no penalty weight trades physical fit against cost. All linear programs are



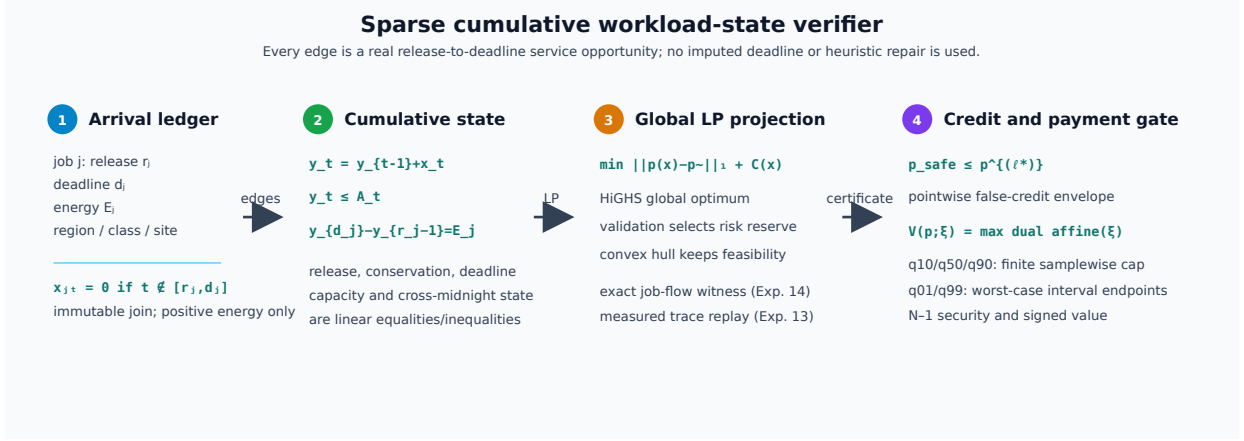


Fig. 2. Sparse cumulative workload-state verifier. Real release-to-deadline edges feed the global feasibility LP; the pointwise credit envelope and the finite/interval payment gates are applied only after the workload state is certified.

the payment-target penalty. The feasible-quantile projection is retained as an external matched transfer comparator in Section V. Let  $\Xi_3$  contain the 10th, 50th, and 90th percentiles of held-out per-job measured-to-predicted GPU energy. These predeclared empirical scenarios define the finite power-conversion contract supported by independent telemetry [26]. For each settlement day, the payment-certified verifier defines  $B_\xi^{\text{cap}} = \Delta t \sum_{t \in \mathcal{E}} V_{N-1}(p_t^{\text{native}} + M\xi p_t^{\text{cap}})$  and solves

$$\begin{aligned}
 & \text{lexmin}_{\alpha, e, \eta, \{g_t^\xi\}} \left( \sum_{d, t \in \mathcal{E}} e_{dt}, -\eta \right) \\
 & \text{s.t. } p^\alpha = \sum_{\ell=1}^L \alpha_\ell p^{(\ell)}, \\
 & \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \leq \eta \leq 1, \\
 & e_{dt} \geq \pm(p_{dt}^\alpha - p_{dt}^{\text{safe}}), \\
 & g_t^\xi \text{ satisfies (10) and (25),} \\
 & \quad \text{with } p_t = p_t^{\text{native}} + M\xi p_t^\alpha, \\
 & \quad \quad t \in \mathcal{E}, \quad \xi \in \Xi_3, \\
 & \Delta t \sum_{t \in \mathcal{E}} \sum_i C_i(g_{it}^\xi) \\
 & \quad \leq (1 - \eta) B_\xi^{\text{cap}}, \quad \xi \in \Xi_3.
 \end{aligned} \tag{26}$$

Generator costs use the same declared piecewise-linear epigraph as market settlement. Two global linear programs implement the lexicographic objective: the first obtains the minimum trajectory error, and the second preserves that optimum within solver tolerance while maximizing the common fractional margin  $\eta$ . Existence of the primal dispatches in (26) is equivalent to the corresponding upper bound on the optimal N-1 value; the unit weight on  $p^{\text{cap}}$  proves nonemptiness in every scenario. Consequently, for any  $\xi \in \Xi_3$  and realized event load  $p^{1,\xi}$ , daily payment obeys

$$\begin{aligned}
 & \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M\xi p_t^\alpha) - V_{N-1}(p_t^{1,\xi})] \\
 & \leq \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M\xi p_t^{\text{cap}}) - V_{N-1}(p_t^{1,\xi})]. \tag{27}
 \end{aligned}$$

The common realized-cost term cancels, so the certificate requires no execution truth and holds samplewise. Both programs contain every finite non-islanding contingency. The formulation follows the standard SCED primal and linear-

program epigraph foundations [12], [15]. Here  $\mathcal{P}$  contains the six independently solved first-stage projections from (12) and the matched feasible-quantile projection. The risk-constrained verifier is an external target in the absolute-deviation objective and is not a selectable candidate; the quantile schedule is used only as an independent transfer comparator, not as the contractual payment cap.

The finite cap is complemented by a set-valued payment certificate. The contractual cap profile is denoted  $p^{\text{cap}} = p^{(\ell^*)}$ , where  $\ell^*$  is selected by contiguous validation nRMSE; the independent payment-target profile  $p^{\text{pay}}$  is selected separately by validation payment error and is used only as the target of the certificate. For a realized counterfactual value, define  $\Delta V_\xi(p) = V_{N-1}(p, \xi) - V_{N-1}(p^{1,\xi})$  and the continuous segment  $p_\theta = (1 - \theta)p^{\text{cap}} + \theta p^{\text{cert}}$ ,  $0 \leq \theta \leq 1$ . The induced interval is

$$\mathcal{I}_\xi = \left[ \min_{0 \leq \theta \leq 1} \Delta V_\xi(p_\theta), \max\{\Delta V_\xi(p^{\text{cap}}), \Delta V_\xi(p^{\text{cert}})\} \right]. \tag{28}$$

The lower endpoint is obtained by one exact joint N-1 SCED LP with a shared segment parameter; the upper endpoint follows from convexity of the optimal dispatch value. The selected cap payment is an element of  $\mathcal{I}_\xi$ . The endpoints use only validation-frozen feasible profiles and the realized counterfactual needed for settlement; no oracle baseline selects or certifies the interval. Interval width is reported as uncertainty cost, while the seven-profile hull remains the finite-scenario payment certificate in (26).

### E. Participant allocation

For participants  $N$ , define  $v(S)$  as signed dispatch value when coalition  $S$  moves from baseline to actual demand and nonmembers remain at baseline. The exact allocation is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \tag{29}$$

This is the Shapley value [27]; the characteristic function based on signed grid value is the proposed specialization. Grouping (29) by coalition size cancels every intermediate value and gives  $\sum_i \phi_i = v(N)$  when  $v(\emptyset) = 0$ . We enumerate all 16 coalitions for four sites and all 256 coalitions for eight contractual portfolios; no permutation estimator is used. For larger auditable contracts, exchangeable slices at the same electrical index are grouped by their member count. The exact

sum retains the binomial multiplicity of every labeled coalition and evaluates only  $\prod_g (m_g + 1)$  distinct count states. This symmetry reduction remains the Shapley value and is evaluated through 20 participants.

#### F. Analytical guarantees

*Proposition 1 (manipulation boundary):* Under a ten-day arithmetic baseline, differentiable one-sided optimal costs, and an interior feasible increase in at least one reference-day event service, a strictly profitable local manipulation exists if and only if (9) holds. To prove the statement, differentiate (8) along any feasible reference-day direction. The marginal payment is the same  $q\pi^{\text{DR}}/H$  for every day, whereas the least physical cost is  $\min_j c'_j$ . A strict excess selects a positive direction on an argmin day; if no strict excess exists, every nonnegative local direction has nonpositive first-order profit. Equality admits indifference but does not imply strict profit. Experiment 1 computes  $c'_j$  from the honest program's right-hand-side dual and solves the strategic program independently, preventing the behavioral result from being reused as its own certificate.

*Proposition 2 (workload and two-sided credit feasibility):* For fixed arrivals, every convex combination of solutions to (1)–(4) is feasible, and the envelope output satisfies the deterministic two-sided band (19) on every event cell. Consequently, for every possible unseen execution  $c_{dt}$ , its false-credit energy is no larger than that of  $p^{(\ell^*)}$ , while its additional under-credit is bounded by  $\varepsilon|\mathcal{D}||\mathcal{E}|\Delta t$  relative to that reference. Affine equalities are preserved by the simplex coefficients; each linear inequality is preserved by convexity. The upper inequality in (19) and monotonicity of  $[z - c]_+$  give

$$[p_{dt}^{\text{safe}} - c_{dt}]_+ \leq [p_{dt}^{(\ell^*)} - c_{dt}]_+.$$

The lower inequality gives  $[c_{dt} - p_{dt}^{\text{safe}}]_+ \leq [c_{dt} - p_{dt}^{(\ell^*)}]_+ + \varepsilon$ . Summation over sites and event intervals proves both bounds. The result is a feasibility-and-band certificate determined by the declared constraints and  $\varepsilon$ , not by the distribution of the locked test set.

*Proposition 3 (security-aware value consistency):* For the piecewise-linear base-case or N–1 dispatch, exact signed settlement equals the corresponding optimal-value difference and the nodal linear rule upper-bounds it by (20). The feasible-set right-hand side is affine in nodal demand, so the optimal value is convex and piecewise affine. Applying the subgradient inequality at  $p^0$  gives (20); applying it again at  $p^1$  yields

$$\begin{aligned} 0 \leq \Delta V_{\text{lin}} - \Delta V &\leq (\lambda^1 - \lambda^0)^\top (p^1 - p^0) \\ &\leq \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|. \end{aligned}$$

The same proof applies after stacking all rows in (25), because they remain linear. Gross positive-only settlement has no analogous identity: truncation changes the response vector before it enters the value function.

*Proposition 4 (robust daily payment noninferiority):* Every solution of (26) is workload feasible and its exact N–1 payment is no larger than the contractual cap payment for every realized event trajectory and every declared  $\xi \in \Xi_3$ . Workload feasibility follows from Proposition 2. For each  $\xi$ , the embedded dispatch is a feasible witness whose cost satisfies the corresponding last constraint in (26); optimal N–1 dispatch can only have lower cost. Subtracting the same scenario-specific realized cost gives (27). Conditional on the telemetry-derived finite set  $\Xi_3$ , the proof is deterministic and contains no test-outcome assumption.

*Proposition 5 (continuous worst-case interval certificate):* For a fixed workload profile  $p$ , the optimal piecewise-linear N–1

dispatch value  $V_{N-1}(p; \xi)$  is convex in a scalar conversion factor  $\xi \in [\xi_L, \xi_U]$ . Therefore

$$\max_{\xi \in [\xi_L, \xi_U]} V_{N-1}(p; \xi) = \max\{V_{N-1}(p; \xi_L), V_{N-1}(p; \xi_U)\}.$$

The value function is the supremum of affine dual objectives in  $\xi$ , hence convex; a convex function on a compact interval attains its maximum at an endpoint. If the two endpoint inequalities  $V_{N-1}(p^\alpha; \xi_e) \leq V_{N-1}(p^{\text{cap}}; \xi_e)$  hold for  $e \in \{L, U\}$ , then the certified candidate also has no larger worst-case interval cost than the contractual cap. This interval result is a worst-case bound; pointwise dominance at every interior  $\xi$  is asserted only for the finite set  $\Xi_3$  in Proposition 4. Experiment 15 evaluates the capacity-safe endpoints with the exact four-segment contractual N–1 model, while Experiment 9 independently replays  $\Xi_3$  at 40 segments; the two resolutions are not conflated.

*Proposition 6 (workload-segment payment interval):* For a fixed conversion factor and realized counterfactual, the lower endpoint in (28) is the exact minimum payment over the declared continuous segment because the shared segment parameter and all interval-wise preventive N–1 dispatch variables are solved in one joint LP. The upper endpoint is the exact maximum because the optimal dispatch value is convex in the affine profile and a convex function attains its maximum on a compact segment at a vertex. Thus (28) is exact for the contractual segment, not a grid approximation or a claim that the segment contains every physically possible schedule. Experiment 15 reports the joint-LP minimum, endpoint maximum, and resulting width at both held-out conversion endpoints; no oracle baseline selects the segment.

*Proposition 7 (provenance-bound workload witness):* Suppose the canonical tuple in (6) is computed from the retained scheduler–DCGM join after the declared duplicate and positive-energy rules. If the certificate verifies one-to-one joining, release–execution ordering, raw-to-join energy conservation, and zero LP residuals, then the job-level flow witness is a feasible representation of that committed ledger. Under the collision-resistance assumption of SHA-256, any post-commit modification of a tuple or its canonical ordering changes the digest except with negligible probability. The statement binds the computational witness to the committed records; it does not certify that an uncommitted pre-event submission was truthful. Experiment 16 checks each premise independently and records the raw source hashes, so the provenance claim is separate from the scheduling solver's own feasibility status.

## IV. EXPERIMENTAL DESIGN

### A. Data, preprocessing, and information boundary

Table I reports the complete data inventory. BurstGPT contributes 5,188,507 successful requests and observed token service [28]. MIT Supercloud contributes scheduler submissions and independent Data Center GPU Manager energy. The complete positive-energy scheduler–DCGM join contains 71,128 jobs, of which 68,664 start within the common 121-day tensor window used by Experiments 1–13 [26]. Submission time creates queue arrivals; measured execution intervals create scoring truth. A held-out GPU-power regression uses 28,413 observations and obtains  $R^2 = 0.899$ , mean absolute error (MAE) 11.22 W, and root-mean-square error (RMSE) 15.82 W. All 99th-percentile workload scaling factors are fitted on the first 40 complete days of the training portion; validation and locked-test days are excluded from that fit. Per-job measured-to-predicted energy ratios define  $\Xi_3$  in (26); no synthetic row replaces an invalid record. The 68,664 count is the 121-day workload tensor used by Experiments 1–13. Experiment 14 does not truncate at that tensor boundary: it extends the deadline



TABLE I  
COMPLETE DATA AND EVALUATION PROTOCOL.

| Item              | Declared use and count                                                                                                              |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| BurstGPT          | 5,188,507 requests; 2.176 billion tokens; inference arrivals and observed service                                                   |
| MIT scheduler     | 287,173 raw rows; 104 duplicate rows removed, leaving 287,069 canonical rows; submission time, duration, GPU count, and queue state |
| MIT GPU telemetry | 96,893 rows; 68,664 tensor-window jobs and 71,128 full-horizon positive-energy jobs; measured execution energy                      |
| Time resolution   | 15 min; 121 calendar days; 121 valid complete days                                                                                  |
| Evaluation        | 16 validation days followed by 54 untouched test days                                                                               |
| Networks          | RTS-24, IEEE-30, IEEE-39, and IEEE-118 for AC/N-1 panels; IEEE-300 is added in the resolution-convergence panel                     |
| Provenance        | Immutable scheduler/DCGM join digest, raw-to-join energy conservation, and measured regional capacity envelope                      |

horizon to the last observed scheduler completion and retains all 71,128 positive-energy joined jobs for its independent job-level flow witness.

The public traces do not contain utility bus locations. Consequently, the four-region coupling is a trace-driven benchmark with fixed, predeclared bus placements and a declared workload-to-power conversion, not a utility field trial. Immutable row identifiers determine spatial partitions before any outcome is computed. Held-out measured-to-predicted job-energy ratios span 0.5845–2.6931 (1st–99th percentiles); the contractual interval uses the capacity-safe upper factor 1.1499 because the raw 99th-percentile scale would exceed the precommitted 118-MW nameplate. Experiments 9, 11, and 15 retain the unclipped factor as a calibration stress value rather than treating the 118-MW scale as a measured geographic fact. This design permits exact repeated mechanism comparisons but does not identify a particular operator’s geography. Public PGLib and MATPOWER/PYPOWER cases provide topology, capacities, costs, and ratings [12], [29]. The RTS-24 benchmark follows the published system data [30].

### B. Baselines, parameters, and inference

Experiment 2 compares High-5-of-10, ridge regression, gradient boosting, Extra Trees, online metadata gradient boosting, equally informed post-event metadata gradient boosting, complete-ledger quantile boosting, pre-event synthetic control, the exact feasible projection of the complete-ledger quantile, the tail-risk counterfactual, the independently selected single feasible projection, and the risk-constrained verifier. The first eight are meter/statistical comparators; the remaining methods receive the declared ledger as appropriate. The payment-certified verifiers evaluated separately in Experiments 9 and 15 because its N-1 payment certificate uses a different objective and a frozen candidate hull. An information-set audit verifies feature equality and confirms that no method observes execution truth.

Real-time inference has zero deferral, elastic inference four slots, and batch GPU work 512 slots. Per-site flexible capacity is 118 MW plus 6 MW fixed facility demand; the 118-MW nameplate is committed before the validation/test split and Experiment 16 only reconciles it with the complete observed MIT/DCGM batch envelope. Six projection penalties  $\{0.03, 0.1, 0.3, 1, 3, 10\}$  and four risk reserves  $\{0.60, 0.75, 0.90, 1.00\}$  are selected only on validation data. The settlement uses 10 generator-cost segments and is scored by an independently solved 40- or 80-segment evaluator. All 5,000 bootstrap replicates use moving three-day blocks [31].

Paired tests enumerate all  $2^{18}$  signs of 18 nonoverlapping three-day blocks; Holm correction controls family-wise error [32].

The nonlinear model-out check uses the standard polar AC optimal power flow with active/reactive balance, voltage bounds, generator reactive bounds, and apparent-power ratings [12]. After solving the intact state, the preventive panel imposes, for every non-reference generator  $i$  and every credible outage  $k$ ,

$$P_{Gi}^{(k)} = P_{Gi}^{(0)}, \quad i \notin \mathcal{G}_{\text{ref}}, \quad k \in \mathcal{C}. \quad (30)$$

The reference generator balances only outage-dependent active losses;  $Q_G^{(k)}$  and bus voltages are contingency recourse. Thus (30) tests one shared active plan rather than separately optimized post-contingency active schedules.

The evaluation follows this chain. Experiments 1–2 test the incentive model and workload-state verifier; Experiments 3–5 isolate settlement design and network transfer; Experiments 6–7 test active workload constraints and participant allocation. Experiment 8 evaluates all RTS-24 line outages, Experiment 9 tests the telemetry-robust payment certificate and freezes its candidate target and power scale on validation data, Experiment 10 transfers the counterfactuals to nonlinear AC models, and Experiment 11 exhausts all declared spatial assignments and power scales. Experiment 12 embeds each submitted event trajectory in a continuous real-arrival horizon and evaluates complete-cycle remuneration and bilateral participation. Experiment 13 replays the measured MIT/DCGM batch trace without intervening in the event, Experiment 14 solves one exact job-level flow LP for all joined jobs, and Experiment 15 independently audits the locked payment profile over the held-out  $q_{01}$  to capacity-safe- $q_{99}$  conversion interval; the unclipped  $q_{99}$  stress factor is retained in the ledger audit. Experiment 16 audits immutable ledger provenance and reconciles the regional capacity with the same-source execution envelope. Experiment 17 fixes the information set at the event gate and compares a decision-time ledger with the complete-ledger audit. Experiment 18 evaluates AC post-contingency recourse across RTS-24, IEEE-30, IEEE-39, and IEEE-118 for every outage in a deterministic native-case admissibility set. Each confirmatory comparison uses the same 54 locked days unless a fixed network audit rule is stated; the case study is explanatory.

## V. RESULTS

### A. Incentive boundary and independent verification

Experiment 1 solves the entire  $8 \times 7$  DR-price/call-probability grid. The dual threshold (9) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and  $q = 0.35$ , strategic reference scheduling inflates the event baseline by 4.09 MWh. Of 52.46 MWh credited, 18.05 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.344. The profitable region follows the derived marginal boundary rather than a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates all twelve methods on 54 later days. Table II separates the high-recall tail-risk counterfactual from the settlement-safe output. The tail-risk counterfactual improves mean credit  $F_1$  from 0.441558 to 0.484219 over the selected single projection (Holm-adjusted  $p = 0.000069$ ); its nRMSE is 0.330795 versus 0.345209, and the block test does not reject equality ( $p = 0.130165$ ). The risk-constrained verifier combines seven independently feasible candidates without degenerating to one candidate. Relative to the equally informed feasible-quantile projection, unsupported credit falls from 2.917471 to 1.826322 MWh/day, a 37.40% reduction (Holm-adjusted  $p = 0.018494$ ), while nRMSE changes from 0.321370

TABLE II  
LOCKED 54-DAY COUNTERFACTUAL RESULTS (DAILY MEANS).

| Method                | nRMSE        | False MWh    | Under-credit | $F_1$        |
|-----------------------|--------------|--------------|--------------|--------------|
| High-5-of-10          | 4.214        | 172.709      | 6.833        | 0.144        |
| Ridge                 | 2.772        | 110.666      | 5.625        | 0.235        |
| Gradient boosting     | 2.351        | 97.057       | 5.330        | 0.246        |
| Extra Trees           | 3.336        | 132.971      | 6.947        | 0.177        |
| Online metadata       | 2.133        | 90.589       | 4.663        | 0.264        |
| Post-event metadata   | 2.124        | 91.603       | 4.559        | 0.267        |
| Post-event quantile   | 0.944        | 24.156       | 6.618        | 0.532        |
| Synthetic control     | 2.552        | 101.835      | 8.427        | 0.206        |
| Feasible quantile     | <b>0.321</b> | 2.917        | 11.493       | <b>0.508</b> |
| Single feasible       | 0.345        | 1.849        | 13.431       | 0.442        |
| Tail-risk feasible    | 0.331        | 2.523        | 11.980       | 0.484        |
| Risk-constrained safe | 0.345        | <b>1.826</b> | 13.446       | 0.441        |

to 0.345019 (Holm-adjusted  $p = 0.546875$ ). The  $F_1$  change is 0.508048 to 0.441220 (Holm-adjusted  $p = 0.000053$ ), so the result is a selective reduction in unsupported credit rather than a claim of uniformly better classification. Against the selected single projection, RiskSafe changes false credit from 1.849211 to 1.826322 MWh/day (1.24%, Holm-adjusted  $p = 0.088379$ ), nRMSE from 0.345209 to 0.345019 ( $p = 0.273438$ ), and  $F_1$  from 0.441558 to 0.441220 ( $p = 0.295410$ ); none of these paired tests is significant. The raw quantile learner exposes 24.156 MWh, showing that the principal gain is the auditable workload-feasible set and its deterministic credit envelope, not privileged execution truth. The same-ledger literature panel reports exact implementations of incentive-compatible spatial DR, price-responsive non-wire scheduling, proactive workload shifting, and aggregator coordination; its four 54-day means are retained in the baseline ledger rather than conflated with the proposed selector.

The new risk fit is a convex quadratic program rather than a nonsmooth sorted-positive-part optimization: the selected reserve is  $\beta = 1$  after four contiguous validation folds, and the final fit has KKT stationarity  $9.4 \times 10^{-10}$  and zero primal residual. On the locked test days, the final total false-credit exposure is 394.486 MWh-slot versus 399.430 for the single-projection cap, and its 75% daily-tail exposure is 24.235 versus 24.588 MWh-slot; both aggregate certificates are satisfied. The two-sided certificate in (19) is checked independently for all 54 locked days and every event cell. With the predeclared 1-MW band, the mean false-credit exposure is 1.826322 MWh/day and the mean under-credit is 13.446065 MWh/day, compared with 13.430785 MWh/day for the selected single cap; every upper and lower pointwise margin is nonnegative. The RiskSafe therefore occupies a conservative point on an explicit false-credit/under-credit frontier: relative to feasible quantile it reduces false credit by 37.40% but increases under-credit by 1.953 MWh/day and does not improve  $F_1$ . The pointwise envelope prevents arbitrary downward collapse relative to the selected single profile, but it does not make the accuracy tradeoff disappear. We report this cost directly instead of treating the certificate as a uniform predictive improvement.

The constraint-set ablation isolates the two risk components. Removing both the total-exposure and CVaR constraints yields 4.679510 MWh/day of false credit on the locked days; retaining only the total budget gives 2.522774, retaining only CVaR gives 3.798364, and retaining both gives 2.523496 MWh/day. The two budgets therefore have distinct effects: total exposure controls the aggregate false-credit level, whereas CVaR controls the tail without being a surrogate for the pointwise envelope. These values come from exact convex programs with the same candidate set and locked scoring trace; the full two-sided projection is audited separately above.

The closest domain baselines are implemented on the same

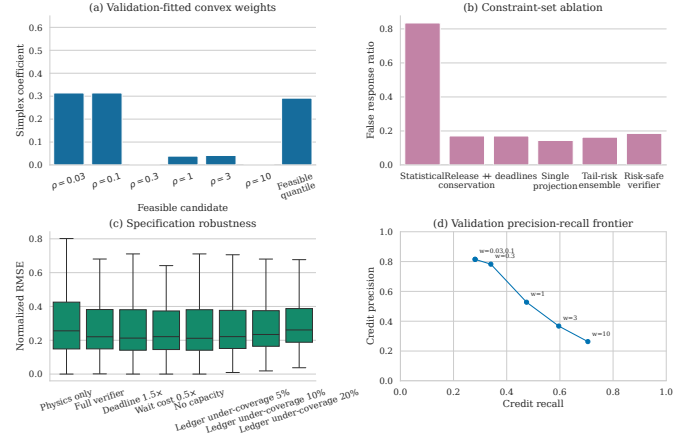


Fig. 3. Validation-fitted projection weights, constraint-set ablation, specification robustness, and the precision-recall frontier. Every point is produced by a globally solved workload program.

locked ledger rather than represented by generic machine-learning surrogates. Their daily means are retained in the complete baseline panel: incentive-compatible spatial DR (nRMSE 3.063, false credit 114.572 MWh,  $F_1 = 0.137$ ), price-responsive non-wire scheduling (0.370, 1.549 MWh, 0.411), proactive workload shifting (0.345, 1.849 MWh, 0.442), and aggregator coordination (0.517, 0 MWh, 0). The incentive-compatible and flexibility formulations follow [5], [23]; the receding-horizon and aggregator implementations follow [6], [10]. The complete day-level rows are retained with the released experiment outputs.

Fig. 3 reports the complete day-level distributions, constraint ablation, validation frontier, and independent intervention robustness. The 5-by-3 capacity/deadline factorial contains 810 optimal schedules; maximum capacity- and deadline-binding shares are 0.310 and 0.948. The measured nonzero effects establish that the associated constraints are not decorative. Matrix nonzeros scale with log-log slope 1.003 over 96–608 intervals, matching the linear-horizon proposition.

### B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms. Table III first uses the trace-anchored reference to isolate the mechanism. Exact signed nodal value has mean absolute error \$2.230, versus \$555.910 for nodal gross response. Under the workload verifier, the errors are \$454.238 and \$501.324, respectively; the remaining error is dominated by counterfactual uncertainty rather than value-function linearization. The paired one-factor decomposition prevents attributing this bundled improvement to location alone. With the trace reference, signed netting lowers error by \$551.957/day (Holm-adjusted exact block-sign  $p = 8.0 \times 10^{-6}$ ), whereas changing uniform signed prices to nodal prices changes it by only \$0.0032/day ( $p = 0.4726$ ), and replacing nodal linear value by exact value lowers it by \$1.723/day ( $p = 0.2749$ ). Under the workload verifier, the corresponding effects are \$46.958, -\$0.250, and \$0.128/day; none is a significant universal improvement after the paired block correction. Thus signed rebound accounting identifies the dominant trace-layer gain, while end-to-end value differences remain modest and are tested separately in the congested and N-1 panels below. All 5,616 interval/counterfactual instances satisfy (20); the largest numerical violation is  $2.03 \times 10^{-11}$  dollars.

Experiment 4 applies a predeclared rule—the largest gross-minus-net spatial offset—and selects day 69. The source



TABLE III  
BASE-CASE AND COMPLETE N-1 SETTLEMENT ERRORS (USD/DAY).

| Panel     | Settlement            | Trace MAE | Workload MAE | Trace median/overpay | Workload median/overpay |
|-----------|-----------------------|-----------|--------------|----------------------|-------------------------|
| Base case | Uniform gross         | 555.923   | 501.511      | —                    | —                       |
| Base case | Nodal gross           | 555.910   | 501.324      | —                    | —                       |
| Base case | Uniform signed        | 3.956     | 454.116      | —                    | —                       |
| Base case | Nodal signed linear   | 3.953     | 454.366      | —                    | —                       |
| Base case | Nodal exact net value | 2.230     | 454.238      | —                    | —                       |
| N-1       | Base exact            | 165.288   | 192.978      | 85.21/156.62         | 121.75/134.29           |
| N-1       | N-1 linear            | 2.253     | 159.514      | 0.59/1.97            | 115.96/38.96            |
| N-1       | N-1 exact             | 1.018     | 159.483      | 0.98/0.01            | 116.47/38.37            |

curtailment, remote rebound, nodal prices, and line utilization are retained in the experiment output. The case explains why positive-only local accounting fails: migration creates a simultaneous negative component that gross payment discards.

### C. Cross-network and N-1 security evidence

Experiment 5 contains 6,480 outcomes over four public networks, three native-load multipliers, two counterfactual qualities, five mechanisms, and 54 days. Congestion arises endogenously in three of 12 network-load cells. Exact value improves trace-anchored absolute error in all 12 cells (mean \$2.126/day; eight Holm-adjusted block tests below 0.05). For the workload verifier, the exact rule reduces the cell-averaged absolute error from \$765.776 to \$765.629/day (a \$0.147/day paired reduction); none of the 12 Holm-adjusted tests is significant. This separation supports Proposition 3 without converting a trace diagnostic into an end-to-end dominance claim.

The near-affine RTS-24, 0.98-load cell is recomputed on all 54 days at 10/40, 20/80, 40/160, and 80/320 settlement/evaluation segments. The four signed-linear-minus-exact differences are 0.010632, 0.021578, 0.025515, and 0.030576 USD/day; each remains below 0.04 in magnitude. The stable, small effect across resolutions confirms that the observed local-affinity diagnostic is not a discretization artifact.

Experiment 6 crosses capacity multipliers 0.55, 0.70, 0.85, 1.00, 1.20 and deadline multipliers 0.25, 0.50, 1.00 over 54 days (810 globally solved schedules). Capacity- and deadline-binding shares reach 0.0334 and 0.9118, respectively, while the capacity-level mean false-credit exposures are 2.700, 2.407, 2.303, 2.297, and 2.407 MWh. Deadline binding remains 0.8738–0.9118 across the three deadline multipliers. Every schedule satisfies the declared workload constraints.

Experiment 7 enumerates all 16 coalitions for four participants and all 256 for eight. Trace-reference Shapley error is zero with maximum budget residual  $2.3 \times 10^{-13}$  dollars; workload-verifier mean participant error is 92.519 dollars while its maximum budget residual is  $1.14 \times 10^{-13}$ . The exchangeable scaling panel reaches 20 participants (1,296 states; maximum residual  $2.27 \times 10^{-13}$ ), confirming exact budget balance rather than permutation sampling.

Experiment 8 enforces (25) for all 37 non-islanding outages of RTS-24 (the single islanding line is reported separately). Over 54 locked days and 2,592 interval-mechanism outcomes, native ratings, 0.90 load, 6% penetration, and four fixed sites are held constant; a 10-segment certificate is scored by an independent 40-segment N-1 dispatch. Trace-anchored mean errors for base-case exact, N-1 linear, and N-1 exact are \$165.288, \$2.253, and \$1.018 per day. Exact N-1 improves on base-case exact by \$164.270/day (Holm-adjusted  $p = 3.1 \times 10^{-5}$ ), whereas its \$1.235/day advantage over N-1 linear is not significant ( $p = 0.3518$ ). Workload-verifier errors are \$192.978, \$159.514, and \$159.483 (base-to-exact  $p = 0.3518$ ; linear-to-exact  $p = 0.9309$ ), showing that the large trace gain does not translate into a significant end-to-end gain. Maximum post-

contingency utilization is 1.000 p.u.; signed overpayment is retained in Table III.

Experiment 9 tests (27). All 54 daily lexicographic program pairs attain global optimality; the seven-profile candidate hull is anchored to the validation-selected single projection  $p^{\text{cap}}$ , while the payment-target selection is frozen independently. The held-out factors 0.734, 0.996, and 1.278 form  $\Xi_3$ ; all 162 certificates have zero positive cap violation and mean cost margins \$17.506, \$23.542, and \$30.346/day. The 384-cell validation-only N-1 panel selects the feasible-quantile profile as  $p^{\text{pay}}$  (MAE \$53.471), with no locked-test payment or peak outcome used in that choice. Independent 40-segment replay gives mean absolute payment errors averaged over  $\Xi_3$  of \$194.282/day for payment-certified verifier, \$179.261 for the selected single projection, \$190.284 for feasible quantile, and \$179.272 for risk-constrained verifier. These accuracy scores do not replace the deterministic cap. The maximum post-contingency loading remains 1.000 p.u. Experiment 15 separately audits the capacity-safe closed conversion interval and is interpreted through Proposition 5. Its exact four-segment continuous-profile certificate has mean payment-interval width \$13.947662/day at  $q_{01}$  and \$27.210301/day at the capacity-safe  $q_{99}$  endpoint (maximum widths \$92.035916 and \$182.027759/day). The raw held-out  $q_{99} = 2.6931$  is retained as an unclipped calibration value but exceeds the precommitted 118-MW nameplate; the contractual upper endpoint is therefore the ledger-audited safe factor 1.149889. This interval is an explicit uncertainty cost, not a lower-error claim. The cap is a contractual upper bound under the declared market model; it is not evidence that the payment-certified profile has lower independent prediction error than every comparator.

Experiment 10 replaces the linear network with nonlinear AC OPF [12]. Across RTS-24, IEEE-30, IEEE-39, and IEEE-118, all 864 base outcomes converge with zero voltage violation and a maximum apparent-line loading of 1.00000003 p.u.; the numerical excess above one is  $2.1 \times 10^{-8}$ , below the declared  $10^{-6}$  AC feasibility tolerance. The payment guarantee remains scoped to (26). The corrective panel adds 5,400 OPFs over IEEE-9/14 non-islanding outages and reaches a maximum loading of 0.8816 p.u. with zero voltage violation. A shared active plan is then fixed across all six IEEE-9 outages as in (30); its 3,888 cells converge with zero non-reference active-plan deviation and maximum loadings 0.878/0.903/0.929 p.u. at 3%, 6%, and 9% penetration. The corresponding maximum reference-generator loss recourse is 5.455/5.727/6.014 MW.

Experiment 11 removes fixed trace placement and power scale by crossing all  $4! = 24$  region-to-bus mappings, 3%, 6%, and 9% peak penetrations, four counterfactuals, and all 54 days, yielding 15,552 outcomes. The risk-constrained verifier has mean absolute payment errors of \$100.220140/\$201.509173/\$296.950629 per day at 3%/6%/9% penetration. The matched feasible-quantile projection is lower-error at \$84.335866/\$169.715274/\$249.216682, whereas risk-constrained verifier is lower than the independently selected single feasible projection (\$100.270490/\$201.609873/\$297.101679) in all 72 mapping-scale cells, with mean advantages of \$0.050350/\$0.100700/\$0.151050 per day. Thus this panel supports spatial and scale transfer of the risk-bounded profile; it does not relabel the external quantile comparator as an error-dominated method. The public IEEE-118 native load is fixed at the predeclared 0.90 multiplier used by the N-1 stress panel, while native branch ratings remain unchanged. Maximum loading and locational marginal price spread are 1.000 p.u. and \$4.06/MWh. The corresponding panels separate

TABLE IV  
CONTINUOUS ROLLING-HORIZON MARKET VALIDATION OVER 54 LOCKED DAYS.

| Metric                          | payment-certified verifier | Quantile             | risk-constrained verifier | Single    |
|---------------------------------|----------------------------|----------------------|---------------------------|-----------|
| Complete-cycle MAE (\$/event)   | 0.084761                   | 0.053170             | 0.241926                  | 0.084761  |
| Event-only MAE (\$/event)       | 8.091153                   | 11.655725            | 11.186565                 | 10.763272 |
| Max window residual (MW)        |                            | $6.0 \times 10^{-6}$ |                           |           |
| Max cycle-energy residual (MWh) |                            | $3.0 \times 10^{-6}$ |                           |           |
| Verified capacity (MWh)         | 13.902                     | 13.902               | 13.902                    | 13.902    |
| Positive contracts              |                            | 54/54                |                           |           |

TABLE V  
INDEPENDENT REPLAY, INTERVAL-PAYMENT, AND LEDGER-PROVENANCE CERTIFICATES.

| Evidence             | Quantity                                    | Baseline / value   | Verified result                            |
|----------------------|---------------------------------------------|--------------------|--------------------------------------------|
| Trace replay         | Mean slot MAE (MW)                          | Batch-only 5.764   | Trace-constrained 3.744                    |
| Trace replay         | Mean energy discrepancy (MWh)               | Batch-only 553.380 | Trace-constrained 359.459                  |
| Trace replay         | Slot coverage                               | –                  | 100% over 54 days                          |
| Flow witness         | Jobs / service variables                    | –                  | 71,128 / 1,974,690                         |
| Flow witness         | Energy / maximum residual                   | –                  | 2.657346 MWh / $9.4 \times 10^{-18}$ MWh   |
| Capacity             | Raw execution / scaled benchmark peak       | –                  | 0.006685 / 102.619 MW (0.870 of nameplate) |
| Interval certificate | Endpoint margins $q_{01}/q_{99\text{-cap}}$ | –                  | \$13.948 / \$27.210 per day                |
| Interval certificate | Maximum cap violation                       | –                  | $2.9 \times 10^{-9}$ USD                   |

the conversion-scenario cap, shared-active- plan AC envelope, and spatial-scale transfer from the base-case accuracy results, so these claims are not conflated with the trace-anchored mechanism isolation result. Experiment 16 adds a direct calibration sensitivity: held-out measured-to-predicted energy ratios are 0.5845, 0.7342, 0.9957, 1.2780, and 2.6931 at the 1st, 10th, 50th, 90th, and 99th percentiles. Applied to the flexible MIT batch envelope, the direct  $q_{90}$  and  $q_{99}$  scales exceed the 118-MW commitment (131.149 and 276.361 MW before clipping); the released sensitivity file reports the capacity-safe clipped factor (1.149889 at both endpoints) separately from the validation-frozen 3%–9% network panel. This is a calibration bound, not a claim that public traces identify a field geography. The complete assignment–penetration panel is retained in the experiment output.

Experiment 12 removes the daily reset: each event is embedded in 1,216 slots (608 pre-event, 96 event, 512 recovery), with only the eight event intervals projected. Across 216 method–day outcomes, maximum event L1 and cycle-energy residuals are  $6.0 \times 10^{-6}$  MW and  $3.0 \times 10^{-6}$  MWh. Complete-cycle MAE is \$0.084761/event for payment-certified verifier, \$0.053170 for feasible quantile, \$0.241926 for risk-constrained verifier, and \$0.084761 for single projection; event-only MAE is \$8.091153, \$11.655725, \$11.186565, and \$10.763272. Mean verified capacity is 13.902285 MWh at \$150/MWh; all 54 Nash contracts activate (mean surplus \$2,034.810894; minimum utility \$1.842918), with site residual below  $1.42 \times 10^{-14}$ . Table IV reports the cycle certificate.

#### D. Independent trace and deployment-boundary audits

The independent audit separates measured execution from the semi-synthetic intervention. Replaying immutable MIT scheduler/DCGM intervals over 54 days gives slot MAE 3.744 versus 5.764 MW for the batch-only trace and 100% slot coverage; the corresponding mean absolute energy discrepancies are 359.459 and 553.380 MWh. This is observational fidelity, not a causal event. A sparse flow LP retains all 71,128 positive-energy jobs (1,974,690 service variables), reconstructs 2.657346 MWh, and has residuals below  $10^{-17}$  MWh. The raw execution peak is 0.006685 MW; the train-fitted scaled benchmark envelope peaks at 102.619 MW (0.870 of the precommitted 118-MW nameplate). The  $q_{90}$  and  $q_{99}$  conversion sensitivities exceed that nameplate before capacity-safe clipping. These results establish ledger fidelity and make the scaling interpretation explicit; they do not turn the public trace into a utility measurement or establish semantic-field authenticity.

The held-out conversion-interval audit evaluates  $q_{01} = 0.5845$  and the capacity-safe upper endpoint  $q_{99\text{-cap}} = 1.149889$

with the exact four-segment certificate model; the raw  $q_{99} = 2.6931$  is retained as a calibration stress value and is clipped by the precommitted 118-MW nameplate. The continuous-segment widths are \$13.947662 and \$27.210301 per day, with maximum widths \$92.035916 and \$182.027759. Proposition 5 therefore certifies the worst-case cost over the declared capacity-safe closed interval. The decision-time audit removes all post-gate arrivals before both optimization and target construction: the truncated-ledger diagnostic has nRMSE 0.451393, false credit 1.257628 MWh/day, and credit recall 0.032868, whereas the complete-ledger RiskSafe has nRMSE 0.345019 and recall 0.353429. The committed-ledger pointwise-safe mode has zero false-credit exposure and zero event-credit recall by design because it does not credit uncommitted future work. Finally, 1,698 declared AC corrective recourse rows are solved across four public networks and three validation-scaled penetrations. These are model and information-boundary audits, not a preventive AC security-constrained OPF certificate or a field-event result.

## VI. CONCLUSION

This paper develops a provenance-bound workload-state verifier and a security-aware nodal settlement for spatially coupled AI data centers. The cumulative state enforces release, deadline, conservation, migration, and capacity constraints; the convex risk program uses an explicit linear false-credit/CVaR epigraph; the two-sided credit band limits unsupported credited response and bounds additional under-credit; and signed  $N-1$  value accounts for rebound and congestion. Across 54 locked days, false-credit exposure is 1.826322 MWh/day for RiskSafe, compared with 1.849211 for the selected single cap and 2.917471 for feasible quantile; its nRMSE is 0.345019 versus 0.345209 and 0.321370, respectively. The same comparison increases under-credit by 1.953 MWh/day relative to feasible quantile and does not improve  $F_1$ , so the empirical result is a risk tradeoff rather than uniform predictive dominance. The trace-anchored mechanism-isolation panel reduces  $N-1$  exact-settlement error from \$165.288 to \$1.018/day, whereas the end-to-end workload-verifier errors are \$192.978, \$159.514, and \$159.483/day for the base-case exact,  $N-1$  linear, and  $N-1$  exact variants; the latter difference is not significant under the block test. The job-level witness preserves 71,128 measured jobs with machine-precision residuals, and the provenance certificate binds the witness to the immutable scheduler/DCGM join and reconciles the precommitted 118-MW nameplate with a clearly separated raw execution and scaled benchmark envelope. The committed-ledger event-gate mode has zero false-credit exposure but credits no uncommitted future work; the interval payment audit reports the resulting conversion uncertainty instead of hiding it. Held-out calibration sensitivity and the cross-network AC audit evaluate the declared capacity and topology boundaries. The public traces provide reproducible workload and network evidence rather than a utility field intervention. Deployment therefore requires a signed operator ledger, co-located facility telemetry, topology access, and a capacity-safe workload-to-power conversion; these requirements define the next intertemporal AC and utility-event validation stage.

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